ATULKUMAR VISHWAKARMA

Backend Engineer | AI & RAG Systems | Node.js | Laravel | REST APIs | Vector Search
+91 9969368255 • atul.vis4@gmail.com • linkedin.com/in/atulvis4 • github.com/atulvis4

PROFESSIONAL SUMMARY

Backend Engineer with 1.3 years of experience building scalable backend systems and AI-powered applications. Experienced in designing multi-agent RAG architectures, vector search pipelines, and LLM-integrated backend services using Node.js, Redis, and Docker. Proven ability to independently architect and ship production systems including AI assistants, RBAC-based platforms, REST APIs, payment integrations, and cloud storage solutions.

EXPERIENCE

Software Developer | **Cancer Mitr** *Nov 2024 – Present*

- Independently architected and shipped the full backend for OneMi App — a production health platform — using Node.js, Prisma, MySQL, Redis, and Docker; engineered 30+ REST APIs supporting AI workflows, authentication, payments, and user data management.
- Designed a multi-agent RAG system with an Orchestrator Agent and 4 specialized agents (general info, product recommendations, lab report analysis, ticketing); implemented LLM-based inter-agent routing with no keyword matching to improve accuracy and response quality.
- Engineered context-aware AI conversations using a rolling summary architecture with a 15-message immediate window; integrated OpenAI Responses API (GPT-4o, GPT-4o Mini, text-embedding-3-small 1536-dim) and Qdrant vector DB for semantic search.
- Developed complete membership and monetization backend: per-user token logging and budget enforcement, Razorpay payment gateway, cart system, and OTP-based auth with OTPs cached in Redis to reduce DB load.
- Architected RBAC across 5-6 tables for fine-grained access control; migrated file storage from local filesystem to Amazon S3 with a dedicated S3 bucket for centralized application logging.
- Engineered an admin-controlled product seeder pipeline — reads from SQL, generates vector embeddings, and seeds structured data into Qdrant for AI-powered semantic search.
- Developed REST API backends for CancerCost Calculator and Symptom Tracker using Laravel and MySQL.

FullStack Web Developer Intern | **Uponly Technologies** *Jan 2024 – Aug 2024*

- Collaborated on an HRMS Portal (React frontend, PHP/SQL backend) with features for employee management, payroll, and attendance tracking; used Redux Toolkit for state management.
- Contributed to a unified Web Admin Portal integrating HRMS and Lead Management System (LMS) with a focus on security and scalability.
- Worked on the Uponly Pro mobile app backend — built APIs in PHP/SQL and coordinated with frontend developers to integrate them into the mobile application.

Software Developer Intern | **Surya Engineering** *Jul 2023 – Dec 2023*

- Built the company's website frontend using React and Redux Toolkit, ensuring performance, maintainability, and alignment with business requirements.

KEY PROJECTS

OneMi App — Backend (Cancer Mitr)

- Production health-tech platform backend built solo; stack: Node.js, Prisma, MySQL, Redis, Docker, Qdrant, Amazon S3, OpenAI API.
- Multi-agent AI system with orchestrator + specialized agents, context-aware conversations, rolling summaries, and LLM-driven inter-agent delegation.
- Full membership management, token usage tracking, RBAC, Razorpay, OTP auth, Health Locker, and admin seeder pipeline.

CancerCost Calculator & Symptom Tracker (Cancer Mitr)

- RESTful backend APIs built with Laravel (PHP) and MySQL for two patient-facing healthcare tools.

HRMS Portal (Uponly Technologies)

- Full-stack HRMS with employee management, payroll, and attendance; React + Redux Toolkit frontend, PHP + SQL backend.

Uponly Web Admin Portal

- Unified admin platform combining HRMS and LMS for external enterprise clients; React frontend, PHP backend, emphasis on security and scalability.

TECHNICAL SKILLS

Languages	JavaScript (Node.js), PHP, SQL, HTML, CSS
Backend	Node.js, Express, Laravel, Prisma ORM, REST APIs, Backend Architecture
Databases	MySQL, MongoDB, Redis, Qdrant (Vector DB)
AI / RAG	OpenAI API (GPT-4o, GPT-4o Mini, text-embedding-3-small), Multi-Agent RAG, AI Systems, Vector Search, Context Engineering, Rolling Summaries
DevOps / Cloud	Docker, Amazon S3, AWS, Git, GitHub
Frontend	React, Redux Toolkit, TailwindCSS (supporting role)
Tools	Postman, Razorpay, Prisma Migrate, XAMPP

EDUCATION

B.E. in Mechanical Engineering — L. R. Tiwari College of Engineering, Mumbai University
2019 – 2023 | CGPI: 8.29

ACHIEVEMENTS & EXTRA-CURRICULAR

- Silver Medal — State-Level Boxing Championship.
- Head, Designing Club — organized workshops and competitions, trained members in creative and technical design.

LANGUAGES

English • Hindi • Marathi